Arunabha Majumder

Engineering Portfolio | Mechanical Design, Prototyping and Testing

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A short overview of hardware I have designed, built and tested. Each project lists the problem, my own contribution and the result.

1. HYBRID-ACTUATED SHOULDER EXOSKELETON (PHD, PROJECT VIEXO)

Aalborg University, 2023 - present

- **Problem:** Motor-only shoulder exoskeletons need a large motor to carry the full weight of the arm, which costs energy and adds mass.
- **My role:** Concept, mechanical design, prototyping, sensor integration and testing.
- Came up with a hybrid concept: a spring carries most of the gravity load and a small 6 Nm motor adds the rest.
- Designed the parts and assembly in SolidWorks, from hand sketches to detailed models.
- Built and rebuilt the parts with 3D printing, fixing weak parts after each test round.
- Integrated the motor, encoder and force sensors into the frame and built my own test rigs.
- Collected data from 12 participants; worked in a team of 3 on the controller (my part: equations such as the Fourier fit, and GUI behaviour).
- **Result:** More than 25% lower energy use; Best Research Paper Award, IFToMM ISRM 2026.
- **Skills shown:** Concept to tested hardware, mechanism design, mass-conscious design, electromechanical integration, test and iteration.

2. FORCE ESTIMATION ON A 5-BAR PARALLEL ROBOT

Aalborg University, PhD

- **Problem:** Measure the force at the tip of a robot without placing a sensor there.
- **My role:** Estimation method and validation.
- Used the variable-stiffness actuators and the robot Jacobian to estimate the end-effector force.
- Validated the estimate against a reference force sensor; used it to measure human arm impedance in the gait lab.
- **Result:** Within 10% RMS error of the reference sensor.
- **Skills shown:** Kinematics, measurement and validation, test planning.

3. UPPER-LIMB REHABILITATION EXOSKELETON (BACHELOR'S CAPSTONE)

Siddaganga Institute of Technology, 2018 - 2019

- **Problem:** Low-cost device to support arm rehabilitation.
- **My role:** Led the design and build.
- Designed it in CAD, machined the parts, assembled it and tested it with users.
- **Result:** Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.
- **Skills shown:** Design for build, machining, assembly, user testing.

4. STUDENT VEHICLES: GO-KART (2017) AND EFFICYCLE HYBRID VEHICLE (2018)

Siddaganga Institute of Technology

- **Problem:** Build working vehicles with student teams for competitions.
- **My role:** Team member in design, fabrication and assembly.
- Designed, fabricated and assembled a Go-Kart from the ground up.
- Built a human-electric hybrid vehicle for the Efficycle competition.
- **Result:** Working vehicles built and run.
- **Skills shown:** Teamwork, fabrication, hands-on troubleshooting.

DRAWING AND COMMUNICATION

- Diploma in Fine Arts. I sketch concepts by hand before CAD and make technical illustrations and figures for papers, posters and demos. SolidWorks engineering drawings; Certified SolidWorks Associate.
- Published at IEEE CONECCT 2022 and ISRM 2026; taught and supervised Bachelor's students (2023 - 2025).

WHAT I AM STILL BUILDING UP

Space hardware, GD&T to full production standard, PLM systems such as 3DEXPERIENCE, and formal change control (ECO/ECR/ECN). These are areas I want to grow in, and I learn tools quickly.